

Data format "2023_dx_Leaf_Fitting_Tomato_Stress.mat"

The data set is organized in two different structures:

PAR = 300 In this example: 35 = 3 treatments x 5 replicates x between 2-3 samples 525 = number of spectral samples for spectral range between 400 - 800 nm. 369 = number of spectral samples for spectral range between 500 - 780 nm.	<table> <tr> <th>Field ▲</th><th>Value</th></tr> <tr><td>wvl_400_800</td><td>525x1 double</td></tr> <tr><td>wvl_500_780</td><td>369x1 double</td></tr> <tr><td>A_eff_Chla</td><td>369x35 double</td></tr> <tr><td>A_eff_Chlb</td><td>369x35 double</td></tr> <tr><td>A_eff_Carb</td><td>369x35 double</td></tr> <tr><td>A_eff_Xan</td><td>369x35 double</td></tr> <tr><td>A_eff_Anc</td><td>369x35 double</td></tr> <tr><td>fAPAR_Chla</td><td>525x35 double</td></tr> <tr><td>APAR_spc_photons</td><td>525x35 double</td></tr> <tr><td>APAR_photons</td><td>1x35 double</td></tr> <tr><td>FQE</td><td>1x35 double</td></tr> <tr><td>str treatment</td><td>35x1 string</td></tr> <tr><td>str repli</td><td>35x1 string</td></tr> <tr><td>str sample</td><td>35x1 string</td></tr> </table>	Field ▲	Value	wvl_400_800	525x1 double	wvl_500_780	369x1 double	A_eff_Chla	369x35 double	A_eff_Chlb	369x35 double	A_eff_Carb	369x35 double	A_eff_Xan	369x35 double	A_eff_Anc	369x35 double	fAPAR_Chla	525x35 double	APAR_spc_photons	525x35 double	APAR_photons	1x35 double	FQE	1x35 double	str treatment	35x1 string	str repli	35x1 string	str sample	35x1 string
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Where for PAR = 300 and PAR = 1000:

Wvl_400_800 - wavelength range between 400 - 800 nm (nm).

Wvl_500_780 - wavelength range between 500 - 780 nm (nm).

A_eff_Chla - chlorophyll a effective absorbance between 500 - 780 nm (-).

A_eff_Chlb - chlorophyll b effective absorbance between 500 - 780 nm (-).

A_eff_Carb - beta carotene effective absorbance between 500 - 780 nm (-).

A_eff_Xan - xanthophyl effective absorbance between 500 - 780 nm (-).

A_eff_Anc - Anthocyanin effective absorbance between 500 - 780 nm (-).

fAPAR_Chla - spectral signature of the fraction of absorbed photosynthetic active radiation (PAR) by chlorophyll a between 500-700 nm(-).

APAR_spc_photons - spectral signature of the chlorophyll a absorbed photosynthetic active radiation (PAR) between 500-700 nm in photons($\mu\text{mol m}^{-1} \text{s}^{-1}$).

APAR_photons - chlorophyll a absorbed photosynthetic active radiation (PAR) in photons($\mu\text{mol m}^{-1} \text{s}^{-1}$).

fqe - fluorescence quantum yield (photons emitted/photons absorbed).

Treatment - treatment time (c: control, s: water deficit, n: nitrogen deficit)
Repli - replicate number
Sample - sample number || L1x → pigments were analyzed from this sample.

¿How to call structure in MATLAB?

Write the structure name (PAR300, PAR1000) + "." + variable name

Example:

PAR300.wvl

PAR300.PAR

PAR300.Refl_real